

Logic

II

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X

mention an incontestable instance of it; - incontestable, I mean
by any fair mind competent to weigh the evidence. That
admirable genius, ^{Prof.} Mr. Flinders Petrie, found at the ancient
trading town of Naukratis 158 weights on the standard or
standards of Egyptian ket. ~~This set was about 110 to 150~~
The great majority of them are of the most unchangeable
material, basalt or syenite; and to only 20 of them was
it necessary to apply any correction to get their original
~~ing~~ amounts; and these few corrections are mostly small.
Now it is not likely that ^{many of these} ~~they~~ were originally in error by
more than about ^{two} ~~one~~ percent ~~on the average~~; and this
likelihood is supported by ^{the fact that half of these between the lightest} ~~a very strong argument from the~~
~~and the first maximum of frequency also differ from that maximum by~~ ^{1.19 gm}
~~the doctrine of errors too elaborate to be here given. Now~~
making the probable error of a weight ^{one and a half per cent} ~~one half per cent~~. Now
ten percent of them, or sixteen, are on the basis of from
136.8 to 139.0 grains to the ket. Therefore, they must
have been designed to be upon a standard of about 139
grains; and this is supported by the fact that they ^{are} ~~much~~
^{more} ~~most~~ numerous ^{the nearer they} ~~near to that~~ are to that standard, namely

The same result
at the other
end.

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mention an incontestable instance of it; - incontestable, I mean by any fair mind competent to deal with the question. That admirable reasoner, Prof. Flinders Petrie, found at the ancient trading town of Naukratis 158 weights having the Egyptian ket as their unit. The great majority were of the most unchangeable material, so that only one in eight required any correction to bring them to ^{its} ~~their~~ ancient amount; and in the majority of cases the corrections do not exceed 7 per cent. There cannot well have been many weights which did not come within ~~to~~ two per cent of these standards at which they aimed; and this estimate is confirmed by the fact that bisecting the ^{number of weights} difference between the lightest weight and lightest inferred standard, as well as that between the heaviest weight and the heaviest inferred standard, we got in ^{the two} ~~both~~ cases weights differing from those inferred standards by 1.1 and 0.9 grains, or 0.7 of one per cent which should be the probable error of a single weight. I shall notice below a possible objection to this inference. Now notice the distribution of the values as shown in the following table:

Ranging from 137.7 to 138.1, there are 7 weights

" " 138.2 to 138.6 " " 4 "

" " 138.7 to 139.1 " " 7 "

$$\begin{array}{r} 136.8 \\ 138.85 \\ \hline 2.05 \end{array}$$

157

22

$$\begin{array}{r} 150 \\ 138 \\ \hline 12 \\ 6 \end{array}$$

144 g

$$\begin{array}{r} 138.85 \\ 139.9 \\ \hline 1.05 \end{array}$$
$$\begin{array}{r} 148.9 \\ 149.8 \\ \hline 0.9 \end{array}$$

1.1 ~~~~~ 0.9

१७१

42

212
157

[illegible]

24

$$11x + 6(1-x) = 6x + 7(1-x)$$

$$5x + 6 = -x + 7$$

$$6x = 1 \quad 4$$

$$x = \frac{1}{6} \quad 10$$

$$20$$

$$7 \quad 6$$

$$10 \quad 35$$

$$11 \quad 4$$

$$30 \quad 55$$

$$x \cdot 5.6 + (1-x) \cdot 8.5$$

$$x \cdot 8.5 + (1-x) \cdot 9.6$$

$$x \cdot 9.6 + (1-x) \cdot 4.9$$

$$x \cdot 4.9 + (1-x) \cdot 3.8$$

$$x \cdot 1.1 = (1-x) \cdot 4.7$$

$$x \cdot 5.8 = 4.7$$

$$x = \frac{4.7}{5.8} = 0.8$$

$$\frac{2.8}{3.3} = 6.1$$

$$\frac{7.7}{1.8} = 4.2$$

$$1.2$$

$$\frac{2.8}{1.8} = 4.6$$

$$\begin{array}{r} 1 \\ 7 \\ 14 \\ 24 \\ 27 \\ 41 \\ 41 \\ 59 \end{array} \quad \begin{array}{r} 2.0 \\ 3.8 \\ 15.2 \\ 3.5 \\ 15.7 \\ 31.4 \end{array}$$

-2

-1

0

2

4

8

-1

5

4

12

7

5

5

4

$$\frac{3.2}{2.2}$$

$$\frac{4.2}{1.2} = 6.3$$

$$1.8$$

$$\frac{2.1}{2.7} = 2.8$$

$$2.1$$

$$\frac{3.5}{2.1} = 4.9$$

$$1.5$$

$$\frac{2.8}{1.5} = 3.5$$

$$2.1$$

$$\frac{1.4}{1.2} = 2.1$$

$$6$$

$$2.6 + x = 4.3$$

$$x + y = 4.8$$

$$y + 1.7 = 6.0$$

$$x = 1.7$$

$$y = 4.3$$

$$x + y = 4.8$$

$$\begin{array}{r} 0.8 \\ 1.8 \\ 3.6 \\ 4.0 \\ 6.4 \\ 6.2 \\ 10.0 \\ 5.4 \end{array}$$

$$\begin{array}{r} 0.7 \\ 1.7 \\ 3.4 \\ 4.0 \\ 6.1 \\ 6.3 \\ 9.5 \\ 6.1 \\ 5.4 \end{array}$$

$$\begin{array}{r} 4.2 \\ 4.4 \\ 2.7 \\ 2.7 \\ 2.8 \\ 2.8 \\ 1.7 \\ 3.2 \\ 3.2 \\ 3.2 \\ 3.2 \end{array} \quad \begin{array}{r} 4.6 \\ 5.4 \\ 6.0 \\ 8.1 \\ 4.8 \\ 5.8 \\ 4.9 \\ 9.6 \\ 8.5 \\ 5.6 \end{array}$$

$$\begin{array}{r} 3.5 \\ 1.2 \\ 3.5 \\ 3.5 \\ 3.5 \\ 3.5 \\ 3.5 \\ 3.5 \\ 3.5 \\ 3.5 \end{array} \quad \begin{array}{r} 1.2 \\ 3.8 \\ 4.3 \\ 4.7 \\ 6.4 \\ 8.6 \\ 7.1 \\ 3.8 \\ 4.8 \\ 4.6 \\ 2.4 \\ 2.7 \\ 0.9 \end{array}$$

$$\begin{array}{r|l} 1 & 0 & 1.7 & 1 & 4.3 \\ 0 & 0 & & 1 & 4.8 \\ 1 & 1 & 4.8 & & \\ \hline 2 & & & & \end{array}$$

$$\begin{array}{r} 4.3 \\ 8 \\ 34.4 \\ 4.8 \\ 38.4 \\ 7.4 \\ 29.6 \\ 4.3 \\ 33.9 \\ 6.78 \end{array}$$

$$\begin{array}{r} 58 \\ 116 \\ 174 \\ 232 \\ 290 \\ 348 \\ 406 \\ 464 \\ 522 \\ 580 \end{array}$$

$$\begin{array}{r} 5.8 \\ 1.0 \\ 9.8 \\ 4.9 = 4.7 \end{array}$$

$$-x \cdot 0.7 = (1-x) \cdot 4.7$$

$$4x = 4.7$$

$$2x + y = 6.5$$

$$x + 2.4 = 9.1$$

$$y = 3.9$$

$$x + y = 5.2$$

$$y - x = 2.6$$

$$x + y = 4.8$$




0	1.00
1	1.1052
2	1.2214
3	1.3499
4	1.4918
5	1.6489
6	1.8222
7	2.0122
1.00	2.7183
1.50	4.4817
2.00	7.3891
2.50	12.1825
3.	20.0885

¹
 2.7183

.905
.819
.741
.670
<u>.606</u>
497
.368
.223
.135+
.082
.05

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X

Range of apparent
standards

Numbers
of weights

The same
smoother by
0.7 each + 0.3 previous

Separated
into classes

136.7 - 137.1

1

0.7 = 0.7

137.2 - 137.6

2

1.7 = 1.7

137.7 - 138.1

4

3.4 = 3.4

138.2 - 138.6

4

4.0 = 4.0

138.7 - 139.1

7

6.1 = 6.1

139.2 - 139.6

6

6.3 = 6.3

139.7 - 140.1

11

9.5 = 9.5

140.2 - 140.6

4

6.1 = 6.1

140.7 - 141.1

6

5.4 = 5.4

141.2 141.6

4

4.6 = 4.2 + 0.4

It is plain that these cluster about, say, 140 grains as the standard aimed at. But now we come to at least one, more maximum, as shown by the following continuation of the table:

weights ~~are~~ are small, and were therefore used for weighing costly or precious matter, our knowledge of ancient metrology enables us to guess with some confidence that ~~the probable error of a single one~~ about half the weights would differ from their virtual standards by more and about half by less than $\frac{1}{2}$ per cent. or say about $\frac{1}{2}$ ^{$1\frac{2}{3}$} grains on a ket. Now these weights vary over 15 grains in such a way that there ~~is~~ ^{hardly} ~~is~~ ^{are} ~~any~~ ^{are} ~~gaps of many~~ ^{gaps of many} ~~tenths of~~ grains in the whole interval ~~which I do not represent the~~ ^{For were the standards fewer such gaps must appear.} where no weight among the 144 belongs. It is probable, therefore, that ~~there are~~ ^{are} five different kets, - probably the standard of different towns, - are represented here. They thus belong to five different natural classes, - certainly to several; - and yet it is in many cases impossible to say which one of two or even three standards a given balance-weight was intended to copy; that is, to which of the natural classes it belongs. These classes then are merged into one another, without in the least ceasing to be distinct.

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$$\begin{array}{r}
 20) 158 \\
 10) 316 \\
 8) 158 \\
 \hline
 19 \quad 150 \\
 2) 38 \quad 300 \\
 \hline
 133 \frac{2}{3} \\
 2) 266 \\
 \hline
 133 \\
 1 \frac{1}{2} \quad \frac{14}{700} \quad \frac{2}{100}
 \end{array}$$

$$\begin{array}{r}
 1268 \quad - .02 \\
 24 \quad - .04 \\
 268 \quad - .02 \\
 1472 \quad - .03 \\
 \hline
 1400 \quad .10 \\
 28 \quad .08 \\
 \hline
 1400 \quad .18 \\
 1400 \quad .10 \\
 \hline
 1400 \quad .28 \\
 21 \quad .3 \\
 \hline
 721 \quad 103 \\
 9 \quad 10 \\
 \hline
 639 \quad 71 \\
 14 \quad 2 \\
 \hline
 639 \quad 73
 \end{array}$$

On ^{apparent} standard of 136.7 to 137.1 grains, there ^{is} 1 weight

" " " " 137.2 to 137.6 " , There are 2 weights

" " " " 137.7 to 138.1 " , " " 4 ~~weights~~

" " " " 138.2 to 138.6 " , " " 4 ~~weights~~

" " " " 138.7 to 139.1 " , " " 7 "

" " " " 139.2 to 139.6 " " " 6 "

The numbers begin to fall off. But not so fast as they should ^{probably} for now we are approaching another standard. Namely

On apparent standard of 139.7 to 140.1 grains, there are 11 weights

" " " " 140.2 to 140.6 " " " 4 "

We have, at any rate, passed one standard. But now we approach another

On apparent standard of 140.7 to 141.1 grains, there are 6 "

" " " " 141.2 to 141.6 " " " 4 "

" " " " 141.7 to 142.1 " " " 6 "

" " " " 142.2 to 142.6 " " " 6 "

" " " " 142.7 to 143.1 " " " 9 "

" " " " 143.2 to 143.6 " " " 3 "

$$\begin{array}{rcl}
 t & & = 0.9 \\
 t+x & & = 5.6 \\
 x+y & & = 4.9 \\
 y+z & & = 4.6 \\
 z & & = 4.0
 \end{array}$$

$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
 $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\begin{array}{rcl}
 2t+x & = & 6.5 \\
 t+2x+y & = & 10.5 \\
 x+2y+z & = & 9.5 \\
 y+2z & = & 8.6
 \end{array}$$

$$\begin{array}{r}
 19.0 \quad 21.0 \\
 \underline{8.6} \quad \underline{6.5} \\
 10.4 \quad 14.5 \\
 4.1
 \end{array}$$

$$\begin{array}{rcl}
 3x+2y & = & 14.5 \\
 2x+3y & = & 10.4 \\
 \hline
 \frac{1}{2}(x+y) & = & 2.49 \\
 \frac{1}{2}(x-y) & = & 2.05
 \end{array}$$

$$\begin{array}{l}
 x = 4.54 \\
 y = 0.44
 \end{array}$$

$$\begin{array}{r}
 6.5 \\
 \underline{4.54} \\
 1.96 \\
 \hline
 t = 9.8 \\
 z = 4.08
 \end{array}
 \quad
 \begin{array}{r}
 8.6 \\
 \underline{4} \\
 8.1
 \end{array}$$

Logic

$\frac{11}{13}$

X

App. standard

No. weights Smoothed Separated

141.7 - 142.1

6

5.4 = 2.7 + 2.7

142.2 - 142.6

6

6.0 = 2.0 + 4.0

142.7 - 143.1

9

8.1 = 0.7 + 7.4

143.2 - 143.6

3

4.8 = 3.7 + 1.1

143.7 - 144.1

7

5.8 = 3.0 + 2.8

144.2 - 144.6

4

4.9 = 0.4 + 4.5

144.7 - 145.1

12

9.6 = 9.6

145.2 - 145.6

7

8.5 = 8.5

145.7 - 146.1

5

5.6 = 1.0 + 4.6

146.2 - 146.6

5

= 2.4 + 2.6

5.0 = 3.0 + 1.3

146.7 - 147.1

4

4.3 and very likely another at about 142.5 grains

The centre of clustering, or
There must have been a standard at about 145 grains, But

now we come to another

147.2 - 147.6

5

4.7 = 4.7

147.7 - 148.1

7

6.4 = 6.4

148.2 - 148.6

5

5.6 = 5.6

148.7 - 149.1

8

7.1 = 7.1

149.2 - 149.6

2

3.8 = 3.8

149.7 - 150.1

6

4.8 = 4.8

150.2 - 150.6

4

4.6 = 4.6

150.7 - 150.1

2

2.6 = 2.6

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We may be confident that there was another standard at about 149 grains. Here, then, are at least three natural and real classes of weights, aiming respectively at standard of 140, 145, and 149 grains; and it is not unlikely there is another at about $142\frac{1}{2}$ grains. But it is utterly impossible to say, ~~at which~~ without additional information, at ^{which} ~~what~~ standards the intermediate weights were aiming.

I mentioned above that some exceptions might be taken to my manner of ^{evaluating} ~~estimating~~ the probable departures of the weights from their ^{virtual} standards. It will be a slight turning aside from our path to stop to consider these at all; for this illustration is only intended to show how real classes very often may be merged into one another; and that has certainly been shown, whatever the facts of this particular case may be. Still, one likes to perfect an illustration, and especially in a work upon logic, it is well to ~~go~~ make illustrations of reasonings rather too full than too meagre. I will, therefore, go a bit out of the

~~FEB 21 1902~~

FEB 22 1902
Washington's birth day

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way, to see whether my procedure was sound. The objection is, that after the weights were made, some considerable part of them might have been rejected as being too light or too heavy beyond the "tolerance", as we call it, in coinage. The result would be the entire removal of all outside these limits making the descent of the curve at both ends very sudden. Therefore the probable error calculated as I have calculated it, from these slopes would be ^{much} smaller than the true probable error. This seems extremely plausible, ~~yet in~~ ^{for in} upon careful examination it seems to be unsound. ~~In~~ the first place, I will show that ~~the~~ the phenomenon this objection supposes was by no means certain to present ~~itself~~ itself, but that whether it did ~~so~~ or not would depend upon what the process of making the weights was. I will ~~then~~ ^{invariably} show that there would be certain criteria by which we could distinguish whether such a phenomenon were present or not; and finally, by applying these criteria, I will

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show that it is, in fact, not present. To begin with, then, there are three conceivable processes whereby the weights might have been made. ~~But~~ ^{first,} Namely, we may suppose that ~~the~~ the workmen were so expert at their trade that they finished the weights without resorting to a balance; and that after they were so finished, ^{they were all} ~~the bad ones~~ tried and the bad ones were rejected. If, however, a balance was employed, it must have been such a balance as is shown on the Egyptian monuments. Now all the balances shown there are equal-armed balances; but they are of two varieties. For some of them have the centre of gravity of the beam above the point of support, while others, the small number have it below. The latter would oscillate like modern balances; the former would remain with one side down until a greater weight was put into the opposite pan when they would turn until stopped, and remain with the other side down. These latter balances only tell whether the thing weighed is heavier or lighter than the weight tried against

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things and afford little indication that the two are nearly equal. For, ~~the~~ ^{with} an equal weights in each pan, they will rest unmoved in either position, instead of oscillating like a modern balance. The second conceivable way of making a weight, then, would be for the workman to have, at his hand one of these top-heavy balances with the maximum allowable weight in it, and to grind down his stone weight until it ceased to throw the balance over. Subsequently, it would be necessary to ascertain whether it would throw over the lightest allowable weight, or not, if it would have to be thrown away. The third way

Such top heavy balances working automatically are in use in all the mints of the civilized world.

conceivable way of making a weight would be for the workman to have an oscillating balance at hand, and to try from time to time how much the weight he was making needed to be ground down. ^{Whatever method} If the first method were employed, and if the workmen had the skill requisite to render that method convenient, the subsequent examination would result

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in the rejection of but a small percentage of the weights, and this would not sensibly effect the calculated probable error.

~~The same thing.~~
If, however, the workmen were unskillful and a considerable

proportion of the weights were rejected, the light ones would

have to be thrown away ^{for the weights are of stone, without loading;} while the heavy ones would be ground

down ^{again;} ~~and~~ ^{if the first method of manufacture were used,} the result would be that very different pro-

bable errors would be ^{calculated} ~~obtained~~ from the heavy weights and

from the light ones, the latter being much larger. Now in

our calculations from the light weights is only a trifle larger than

from the heavy ones; so that the effect is not serious, if this first

method was used. This is the modern method of coinage, except

that machinery is now used, of course. If ~~either~~ the second or the

~~third~~ ^{one} method were employed and this unskillfulness of

the workmen took the form of timidity, the probable error

^{calculated} ~~calculated~~ from the small weights would be ^{again} much greater than

that from the great ones, while if it took the form of overboldness,

the reverse might be the case. It is possible that there might be

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such a mixture of timidity and rashness among the different workmen that the calculated probable error would be ^{as} much too small when derived from the light as from the heavy weights. But this would be an improbably result. If the workmen were few, one might make his maximum at one ~~place~~ ^{or place} and another at another, and thus there would be an appearance on the whole of different standards being employed. There would in fact be different virtual standards, and the only mathematical conclusion is as to the virtual standards. ^{What the nature of their origination was,} ~~How they arose~~ is another question. But different virtual standards produce different natural classes; so that that would not affect my argument. If the third method of manufacture were used, it is quite incredible that any considerable proportion of the weights could have been rejected at a subsequent inspection; and therefore my method of calculation could not fail. In this argument, I could not use as a premise my conclusion* that there were several virtual standards. Otherwise, their existence would at once show that there was no governmental

* I need not say
that it is ^{and} ~~is~~ Petrie's
before me.

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* I have deposited in the Astor Library a list of three hundred such pounds; and they can be but a small fraction of all there were.

supervision, unless it were so indulgent that it would be without effect on my ^{prob} ~~conclusion~~ calculation of probable errors. Since we know that, before the adoption of the ~~denial~~ ^{metric} system, every town on the continent of Europe had its own pound*, and since Egypt was ~~and~~ an empire loosely bound together, it is altogether probable that ^{some of the} ~~the~~ different ~~standard~~ ^{weights} found in Nubia ~~and~~ were intended to conform to the standards of different towns. But none of these weights carry any mark which might warrant their honesty; and therefore we can hardly avoid supposing that ~~the~~ the buyer must have brought his own weights (if not also his own balance). In this case, it seems likely that some weights would be manufactured for the use of buyers and others for the use of sellers; and some of the virtual standards may perhaps ^{have had their origin} ~~be~~ ^{have} in that. ~~the~~ This is somewhat borne out by a set of weights found by Petrie of which the standard is about 80 grains. There is no documentary mention of such a standard; so that its use must have been restricted.

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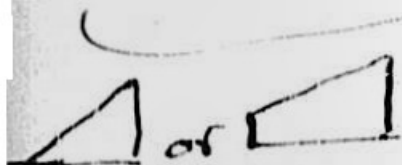
had its own pound. See my article "pound" in the Century Dictionary, which was based on a list of some three hundred pounds known to me, with their values, which list I deposited in the Astor Library. It is, therefore, altogether probable, since Egypt was a loosely connected empire, that every town in Egypt had its own standard of weight; and the kets found at Naukratis ^{are} ^{representative of} probably ~~represent~~ the standards of five important towns. ~~These~~ Virtual standards may, however, originate otherwise. None of these kets carry any mark.

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any definite value, but would simply grind away blindly, trying his weight every time he had ground off about as much as the whole permissible range of variation. Could he guess at that exactly, his error curve would be sectilinear; and so it would if he always ground off the same amount. But since there would be some variability, which would ~~be~~ probably follow the probability curve, his error curve would have a contrary flexure at its lowest part . All this is ~~rather~~ aside from our present purpose; but it will be referred to with interest when we come to study the theory of errors.

I hope this long digression will not have caused the reader to forget that we were engaged in considering some of the consequences of understanding the term "natural," or "real," class to mean a class the existence of whose members is due to a common and peculiar final cause. I had remarked that it is a wide-spread error to suppose that a "final cause" is necessarily a purpose. A purpose is